Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Somatosensory Suppression and Prevention of Post-Flight Re-Entry Disturbances of Posture and Locomotion		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI NNH04ZUU003N Human Health in Space
Start Date:	08/01/2005	**End Date:**	07/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NA00701		
Performance Goal No.:			
Performance Goal Text:			
	Our overall goal is to develop procedures for alleviating post-flight disturbances of posture and locomotion experienced by astronauts. We propose to continue a program of research which has already shown that either non-mechanically supportive contact of the hand with an environmentally fixed surface or pressure cues from a mobile, hand-carried gyroscope stabilizes postural control in a normal 1g environment, a rotating environment, and parabolic flight. Our strategy will be to improve the effectiveness of the portable haptic device and to extend it to situations involving walking. Our original plan was to investigate three avenues of potential improvement: 1) miniaturizing the sensory gyroscope, 2) comparing the gyroscope to a factor vest, and 3) providing subjects training with both devices. At the beginning of this project, we were directed to accelerate aspects of our work which would lead to the development/definition of countermeasures and would enhance collaborative team interactions with other NSBRI balance prosthesis projects. As a result, we spent the first year of the project focusing on research to evaluate the tactile		

Task Description:	sensory encoding and control of passive body sway. We have examined how the latency and pattern of postural stabilization are affected by touch contact of the hand with an environmentally fixed surface during quiet stance and whether position or velocity cues about body sway are keys to establishing stabilization. Our results showed that natural finger contact forces encode body sway velocity, and that the sensed finger forces in turn control the position of the center of foot pressure with a very short latency. These results specified requirements for the motion sensing and stimulus delivery components of a mobile, synthetic system for tactile alleviation of post-flight disturbances of posture and locomotion. Our Year 2 and 3 work aims to evaluate whether body sway and locomotion can be stabilized by a tactor vest device or a few tactors attached to the hand or other body part delivering synthetic tactile cues that provide naturally timed position and/or velocity cues. The results of these studies will provide the basis for designing apparatus and procedures for suppressing re-entry disturbances after spaceflight and hastening sensory- motor readaptation to 1g, in the context of ISS, lunar and Mars missions. This research is most relevant to Risk #14, “Impaired Sensory-Motor Capability to Perform Operational Tasks After Landing and Throughout Re-Adaptation” and Risk #13 “Impaired Sensory-Motor Capability to Perform Operational Tasks During Flight, Entry, and Landing” in the NASA Bioastronautics Roadmap. Continuation of this project will advance the research from the Countermeasure Readiness Level of 4 to a level of 5-6 and will result in a Technology Readiness Level of 3-4.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The haptic cueing theory and technology which we develop in this project can potentially serve as mobility aids to people with neuromotor gait/balance disorders and as a spatial disorientation aid for pilots. We currently have AFOSR funding to develop pilot aids, and we have applied for NIH funding for clinical applications.
Task Progress:	1. Tests were performed on tactors taken from the vibrotactile vest manufactured by TNO Defence, The Netherlands. These are encapsulated DC motors with asymmetric weights on the shaft. Tactors were mounted on the skin, and an accelerometer was mounted on the tactor while it was being driven by DC voltage waveforms from .05 to 5 Hz. The tactors operate from ~1-5 VDC at 50 mA, with principal frequencies varying monotonically from ~90-300 Hz. Four subjects perceived a montonic relationship between drive voltage and overall apparent intensity. There was no measureable time lag of the envelope of the tactor output (raw accelerometer signal) relative to DC voltage sine waves. In summary, 1) tactor performance will not introduce additional lag into feedback signals during postural sway, and 2) the tactors provide perceptually sufficient variation in stimulus intensity. 2. We compared the medial-lateral (M-L) sway of six subjects standing quietly in a tandem foot posture with eyes closed, with 1) no tactile cues, 2) light (<1N) touch of one fingertip on a real surface, and 3) bilateral vibrotactile stimulation of the palms of both hands proportional to the angular rate of M-L body sway (with 6 msec delay). Each subject performed one pre- and one post-training block which included alternating repetitions of each condition, and a training block of 24 trials with vibrotactile feedback. Pre-training, both precision touch on a real surface and vibrotactile velocity feedback attenuated sway significantly ($p<.05$) relative to the no touch condition. Vibrotactile feedback attenuated sway significantly more post- than pre-training. These results demonstrate that just two tactors supplying feedback about body velocity to the hands are sufficient to stabilize M-L sway during quiet stance. 3. We investigated the relationship between light index finger touch force (FTF) on a stationary surface and quiet stance and rhythmic postural oscillations. Ten healthy subjects alternately stood quietly on a force platform with their feet side by side or rocked fore-aft about their ankles at a comfortable frequency, while lightly touching (<1 N) a stable force plate laterally positioned at hip level. Motion of the center of foot pressure (COP), center of mass (COM) and right index FTF were recorded. High maximum cross-correlation values between FTF and anterior-posterior COM and COP motions were found in both conditions. For quiet standing, FTF led COM by 0.49±0.21 s and lagged COM velocity by 0.30±0.25 s. In the rocking condition, no significant lag or lead was found between TFT and COM or COP and their velocities. The cross-correlation between FTF and COM showed a peak magnitude of 0.62±0.21 at ~0.2 Hz for quiet standing and a lower peak value for rocking, approximately 0.44±0.11 at ~0.315 Hz. Thus, during quiet stance FTF seems to represent a reference variable for the motion of the body, and during rocking this role attenuates.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
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